

H34_16

The equation $y^3 - y^2 + y = x^4 + x + 2$ has no integer solutions

Size $h = 34$, length $l = 12$, identifier E1154 (with y replaced by $-y$)

Theorem 1. *The equation*

$$y^3 - y^2 + y = x^4 + x + 2 \tag{1}$$

has no integer solutions.

The proof is an obstruction coming from the cubic Hilbert reciprocity law over the field $K = \mathbb{Q}(\omega)$, where $\omega^2 + \omega + 1 = 0$. To a hypothetical integer solution we attach the values of 20 auxiliary polynomials $H_0, \dots, H_{19}$ with coefficients in $\mathbb{Z}[\omega]$, and a sum B of cubic Hilbert symbols of these values, which is recorded by a weighted graph. By the Hilbert reciprocity law, the local contributions B_p of all primes p add up to 0 modulo 3. We show that $B_p = 0$ for $p \neq 2, 3$ (Section 4), and that $B_2 = 2$ and $B_3 = 0$ (Section 5). This is a contradiction.

Two standard facts from class field theory are quoted without proof: the explicit formula for the cubic Hilbert symbol at the places not above 3, and the Hilbert reciprocity law. Everything else is proved below, up to explicit finite computations: polynomial identities and resultants, which can be checked by expanding both sides or by polynomial division, and the local computations of Section 5, which are finite enumerations of residue classes.

1 Cubic Hilbert symbols over $\mathbb{Q}(\omega)$

The field. Let $K = \mathbb{Q}(\omega)$. Its ring of integers is $\mathbb{Z}[\omega]$, and every element of K can be written uniquely as $A + B\omega$ with $A, B \in \mathbb{Q}$. We write

$$\langle A, B \rangle = A + B\omega,$$

also when A and B are polynomials in x, y or t with rational coefficients. Then $\langle A, B \rangle \langle C, D \rangle = \langle AC - BD, AD + BC - BD \rangle$, and the norm is $N(A + B\omega) = A^2 - AB + B^2$. The units of $\mathbb{Z}[\omega]$ are $\pm\omega^k$.

Places. The finite places of K correspond to the non-zero prime ideals of $\mathbb{Z}[\omega]$. A prime $p \equiv 1 \pmod{3}$ has two places above it, both with residue field $\mathbb{F}_p$, and a prime $p \equiv 2 \pmod{3}$ is inert, with residue field $\mathbb{F}_{p^2}$. The prime 3 is ramified: $\pi = 1 - \omega$ generates the unique prime ideal above 3, and $\pi^2 = -3\omega$, so $9 = \omega\pi^4$. The only infinite place of K is complex. For a finite place v we denote by K_v the completion of K at v and by $v(a)$ the normalized valuation of $a \in K_v^\times$, and we call a a *unit* at v if $v(a) = 0$. For a rational prime p , we write $v \mid p$ if v is a place above p .

Hilbert symbols. For a place v of K and $a, b \in K_v^\times$, the cubic Hilbert symbol $(a, b)_v$ is a cube root of unity, see [1, Chapter V, §3]. We write

$$(a, b)_v = \omega^{\beta_v(a, b)}, \quad \beta_v(a, b) \in \mathbb{Z}/3\mathbb{Z}.$$

We use the following three facts [1, Chapter V, §3 and Chapter VI, §8].

(H1) $\beta_v(a, bc) = \beta_v(a, b) + \beta_v(a, c)$, $\beta_v(a, b) = -\beta_v(b, a)$, and $\beta_v(a, c^3) = 0$. In particular, $\beta_v(a, b)$ depends only on the classes of a and b in $K_v^\times / K_v^{\times 3}$.

(H2) Let v be a finite place not above 3, let q be the number of elements of its residue field, and let ϖ be a uniformizer at v . Write $a = \varpi^e u$ and $b = \varpi^f w$ with units u and w , and define $\chi(u) \in \mathbb{Z}/3\mathbb{Z}$ by $\bar{u}^{(q-1)/3} = \bar{\omega}^{\chi(u)}$, where the bar denotes the reduction to the residue field. Then

$$\beta_v(a, b) = e\chi(w) - f\chi(u).$$

In particular, $\beta_v(a, b) = 0$ if a and b are both units.

(H3) (Hilbert reciprocity.) For $a, b \in K^\times$, $\beta_v(a, b) = 0$ for all but finitely many places v , and $\sum_v \beta_v(a, b) = 0$, where the sum is over all places of K . At the complex place, $\beta_v(a, b) = 0$.

The general formula in (H2) contains an additional term $ef\chi(-1)$, which vanishes because $-1 = (-1)^3$ is a cube. Formula (H2) fixes the normalization of the symbols; with the opposite sign convention all values β_v change sign, and the proof below is the same.

The place above 2. The prime 2 is inert, the residue field is $\mathbb{F}_4 = \{0, 1, \bar{\omega}, \bar{\omega}^2\}$, and $(q-1)/3 = 1$. Hence, for a unit $u = A + B\omega$, the value $\chi(u)$ is 0, 1 or 2 according as (A, B) is congruent to $(1, 0)$, $(0, 1)$ or $(1, 1)$ modulo 2. By (H2) and (H1), the class of $a = 2^e u$ modulo cubes is described by the pair $(e \bmod 3, \chi(u)) \in (\mathbb{Z}/3\mathbb{Z})^2$, and

$$\beta_v(a, b) = e\chi(w) - f\chi(u) \quad \text{for } a = 2^e u, b = 2^f w. \quad (2)$$

If a is a rational integer, then its unit part is a rational odd number, and $\chi(u) = 0$.

The place above 3.

Lemma 2. *Let v be the place above 3. Every non-zero element of K_v is equal to $\pi^a \omega^b 2^c (1 + 3\omega)^d$ times a cube, for unique $a, b, c, d \in \mathbb{Z}/3\mathbb{Z}$. If $\mathbf{a}$ and $\mathbf{b}$ are the exponent vectors (a, b, c, d) of two elements A and B , then $\beta_v(A, B) = \mathbf{a}^\top M_3 \mathbf{b}$, where*

$$M_3 = \begin{pmatrix} 0 & 0 & 2 & 0 \\ 0 & 0 & 1 & 2 \\ 1 & 2 & 0 & 0 \\ 0 & 1 & 0 & 0 \end{pmatrix} \pmod{3}.$$

Proof. For $\alpha \in \mathbb{Z}[\omega]$, the polynomial $g(s) = s + 3s^2 + 3s^3 - \alpha$ satisfies $g'(s) \equiv 1 \pmod{\pi}$, so by Hensel's lemma it has a root s in the completion of $\mathbb{Z}[\omega]$, and then $(1 + 3s)^3 = 1 + 9(s + 3s^2 + 3s^3) = 1 + 9\alpha$. Hence every unit congruent to 1 modulo 9 is a cube in K_v , and the class of a unit modulo cubes depends only on its residue modulo 9. There are 54 units in $\mathbb{Z}[\omega]/(9)$, and a direct enumeration shows that their cubes are ± 1 modulo 9, and that the 27 products $\omega^b 2^c (1 + 3\omega)^d$ with $b, c, d \in \{0, 1, 2\}$ are pairwise distinct modulo 9, also after multiplication by $-1 = (-1)^3$. Hence they represent all 27 classes of units modulo cubes. Since π^3 is a cube, the exponent of π is the valuation modulo 3, and the first statement follows.

By (H1), it remains to compute β_v on the 16 ordered pairs of the elements $\pi, \omega, 2$ and $1 + 3\omega$. Their norms are 3, 1, 4 and 7, hence, for each such pair, all the symbols at the places not above 2, 3 and 7 vanish by (H2), and, by (H3), β_v at the place above 3 is minus the sum of the values at the place above 2 and at the two places above 7. These places correspond to $\bar{\omega} = 2$ and $\bar{\omega} = 4$ in $\mathbb{F}_7$. The classes (e, χ) of the four elements at these places are

	2	$\bar{\omega} = 2$	$\bar{\omega} = 4$
π	(0, 2)	(0, 0)	(0, 2)
ω	(0, 1)	(0, 2)	(0, 2)
2	(1, 0)	(0, 2)	(0, 1)
$1 + 3\omega$	(0, 2)	(1, 0)	(0, 0)

where, at the places above 7, $\chi(u)$ is defined by $\bar{u}^2 = \bar{\omega}^{\chi(u)}$ in $\mathbb{F}_7$, and 7 is a uniformizer. Formula (H2) and the sum over these three places give M_3 . $\square$

2 The graph

Put $f(x, y) = y^3 - y^2 + y - x^4 - x - 2$, so that (1) is the equation $f(x, y) = 0$. The certificate consists of 20 auxiliary polynomials H_i and constants c_i , listed in Table 1, and of weights $e_{ij} \in \{1, 2\}$ for 58 pairs $i < j$, the *edges*. Put $e_{ij} = 0$ for the other pairs $i < j$. We also use $\rho_{ij} = e_{ij}$ for $i < j$ and $\rho_{ij} = -e_{ji}$ for $i > j$, computed modulo 3; these numbers are listed in Table 2.

Table 1: The auxiliary polynomials H_i and the constants c_i . Here $\langle A, B \rangle = A + B\omega$.

i	H_i	c_i
0	$\langle y + 1, 1 \rangle$	$\langle 4, 0 \rangle$
1	$\langle y - 1, -1 \rangle$	$\langle -4, -4 \rangle$
2	$\langle x, 0 \rangle$	$\langle 0, 4 \rangle$
3	$\langle x + 1, 1 \rangle$	$\langle 0, 2 \rangle$
4	$\langle x + 1, 0 \rangle$	$\langle 4, 0 \rangle$
5	$\langle x, 1 \rangle$	$\langle 0, 2 \rangle$
6	$\langle x, -1 \rangle$	$\langle 2, 0 \rangle$
7	$\langle x - 1, -1 \rangle$	$\langle 1, 0 \rangle$
8	$\langle x + y, 1 \rangle$	$\langle 4, 0 \rangle$
9	$\langle x, y - 2 \rangle$	$\langle 2, 4 \rangle$
10	$\langle -x + y - 2, -x^2 - x - 1 \rangle$	$\langle 2, 0 \rangle$
11	$\langle y - 1, 0 \rangle$	$\langle 0, 1 \rangle$
12	$\langle x - y + 1, 0 \rangle$	$\langle 4, 8 \rangle$
13	$\langle x + y, 0 \rangle$	$\langle 0, 1 \rangle$
14	$\langle y - 2, -1 \rangle$	$\langle 0, 2 \rangle$
15	$\langle y - 1, -x \rangle$	$\langle 6, 6 \rangle$
16	$\langle -x + y, 1 - x \rangle$	$\langle 0, -6 \rangle$
17	$\langle y^2 - y + 1, -xy \rangle$	$\langle 1, 0 \rangle$
18	$\langle xy + x + y^2, x + y - 1 \rangle$	$\langle -1, -1 \rangle$
19	$\langle -x^2 + y^2 + 1, -x^2 - xy \rangle$	$\langle -1, -1 \rangle$

Suppose that (x, y) is an integer solution of (1), and write H_i also for the value $H_i(x, y) \in \mathbb{Z}[\omega]$. We will show in Section 3 that all these values are non-zero. For every place v of K put

$$B_v = \sum_{i < j} e_{ij} \beta_v(H_i, H_j) + \sum_{i=0}^{19} \beta_v(H_i, c_i) \in \mathbb{Z}/3\mathbb{Z}, \quad (3)$$

and, for a rational prime p , put $B_p = \sum_{v|p} B_v$.

Proposition 3. *If all H_i are non-zero, then $B_v = 0$ for all but finitely many places v , and $\sum_p B_p = 0$.*

Proof. Apply (H3) to every pair (H_i, H_j) with $e_{ij} \neq 0$, multiply by e_{ij} , and apply it also to every pair (H_i, c_i) . Adding the results gives $\sum_v B_v = 0$, where the complex place contributes 0. Grouping the finite places above the same rational prime gives the claim. $\square$

3 The auxiliary values are non-zero

Let (x, y) be an integer solution of (1). Since 1 and ω are linearly independent over $\mathbb{Q}$, a value $H_i = \langle A, B \rangle$ vanishes if and only if $A(x, y) = B(x, y) = 0$.

Table 2: The numbers ρ_{ij} . A dot denotes zero; $\rho_{ij} = e_{ij}$ for $i < j$, and $\rho_{ji} = -\rho_{ij} \bmod 3$.

ρ_{ij}	0	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19
0	.	.	.	.	.	.	.	.	1	.	.	.	.	1	.	.	1	2	.	.
1	.	.	2	.	1	2	1	.	.	.	2	.	1	2	.	2	.	2	.	.
2	.	1	.	.	.	.	.	.	.	2	.	.	.	.	2	.	.	.	2	2
3	.	.	.	.	.	.	.	.	.	2	2	1	1	.	.	.	1	.	1	.
4	.	2	.	.	.	.	.	.	.	.	2	.	1	.	2	.	.	.	.	2
5	.	1	.	.	.	.	.	.	2	.	.	2	2	.	.	.	.	.	2	2
6	.	2	.	.	.	.	.	2	2	1	.	.	.	.	.	.	.	.	2	2
7	.	.	.	.	.	.	.	.	.	.	1	.	.	.	.	.	.	1	.	.
8	2	.	.	.	.	1	1	.	.	1	.	.	.	.	.	.	2	2	.	.
9	.	.	1	1	.	.	1	.	2	.	.	.	2	.	2	2	.	.	.	.
10	.	1	.	1	1	.	2	.	.	.	.	1	.	2	.	.	.	2	.	.
11	.	.	.	2	.	1	.	2	.	.	2	.	.	.	.	.	2	.	.	.
12	.	2	.	2	2	1	.	.	.	1	.	.	.	.	.	2	.	.	.	.
13	2	1	.	.	.	.	.	.	.	.	1	.	.	.	.	1	.	.	1	1
14	.	.	1	.	1	.	.	.	.	1	.	.	.	.	.	1	.	.	.	.
15	.	1	.	.	.	.	.	.	.	1	.	.	1	2	2	.	.	1	2	.
16	2	.	.	2	.	.	.	.	1	.	.	1	.	.	.	.	.	.	.	1
17	1	1	.	.	.	.	.	2	1	.	1	.	.	.	.	2	.	.	.	.
18	.	.	1	2	.	1	1	.	.	.	.	.	.	2	.	1	.	.	.	2
19	.	.	1	.	1	1	1	.	.	.	.	.	.	2	.	.	2	.	1	.

- For $i = 0, 1, 3, 5, 6, 7, 8, 14$, the coefficient B of H_i is a non-zero constant, and for $i = 10$ it is $-(x^2 + x + 1) \neq 0$. Hence these H_i do not vanish.
- If H_9 , H_{15} or H_{16} vanished, then the two coefficients would give $(x, y) = (0, 2)$, $(0, 1)$ or $(1, 1)$, respectively. But $f(0, 2) = 4$, $f(0, 1) = -1$ and $f(1, 1) = -3$.
- The polynomials $H_2 = x$, $H_4 = x + 1$, $H_{11} = y - 1$, $H_{12} = x - y + 1$ and $H_{13} = x + y$ have rational coefficients. Substituting $H_i = 0$ into f gives the following polynomials in the remaining coordinate t (namely y for $i = 2, 4, 12$ and x for $i = 11, 13$):

i	$f(x, y)$ on $H_i = 0$
2, 4	$t^3 - t^2 + t - 2$
11	$-t^4 - t - 1$
12	$-t^4 + 5t^3 - 7t^2 + 4t - 2$
13	$-t^4 - t^3 - t^2 - 2t - 2$

An integer root of such a polynomial divides its constant term, so the only candidates are ± 1 for $i = 11$ and $\pm 1, \pm 2$ in the other rows, and substitution shows that none of them is a root.

- The coefficient A of H_{17} is $y^2 - y + 1 > 0$. If $H_{18} = 0$, then $B = x + y - 1 = 0$, and then $A = xy + x + y^2 = 1$. If $H_{19} = 0$, then $B = -x(x + y) = 0$, and then $A = -x^2 + y^2 + 1$ is equal to $y^2 + 1$ or to 1.

Hence no H_i vanishes at an integer solution.

4 The primes $p \neq 2, 3$

Charts. We parametrize the curve $H_i = 0$ by a variable t . For $i = 0, 1, 8, 10, 11, 13, 14, 15, 16$, the polynomial H_i has the form $y - r_i(x)$, and we put $\sigma_i(t) = (t, r_i(t))$. For $i = 2, 3, 4, 5, 6, 7, 9, 12$, it has the form $x - r_i(y)$, and we put $\sigma_i(t) = (r_i(t), t)$. For example, $\sigma_0(t) = (t, -1 - \omega)$ and $\sigma_9(t) = (-(t - 2)\omega, t)$. The last three polynomials define conics:

$$H_{17} = y(y - \omega x - 1) + 1, \quad H_{18} = (y + 1 + \omega)(y + x - 1) + 1, \quad H_{19} = (y + x)(y - (1 + \omega)x) + 1.$$

Lemma 4. Let $H = (y - ax - b)(y - cx - d) - k$ with $a, b, c, d, k \in \mathbb{Z}[\omega]$, and let $\delta = c - a$. Let v be a finite place at which δ and k are units, and let $\bar{x}, \bar{y}$ be elements of the residue field of v with $\bar{H}(\bar{x}, \bar{y}) = 0$. Then $\bar{t} = \bar{y} - \bar{a}\bar{x} - \bar{b}$ is non-zero, and

$$\bar{x} = \frac{\bar{t}^2 + (\bar{b} - \bar{d})\bar{t} - \bar{k}}{\bar{\delta}\bar{t}}, \quad \bar{y} = \frac{\bar{c}\bar{t}^2 + (\bar{b}\bar{c} - \bar{a}\bar{d})\bar{t} - \bar{a}\bar{k}}{\bar{\delta}\bar{t}}.$$

Proof. We have $\bar{t}(\bar{y} - \bar{c}\bar{x} - \bar{d}) = \bar{k} \neq 0$, hence $\bar{t} \neq 0$ and $\bar{y} - \bar{c}\bar{x} - \bar{d} = \bar{k}/\bar{t}$. Subtracting this from $\bar{y} - \bar{a}\bar{x} - \bar{b} = \bar{t}$ gives $\bar{\delta}\bar{x} = \bar{t} - \bar{k}/\bar{t} + \bar{b} - \bar{d}$, which is the first formula, and then $\bar{y} = \bar{t} + \bar{a}\bar{x} + \bar{b}$ gives the second one. $\square$

For H_{17} , H_{18} and H_{19} we have $k = -1$, and δ is ω , -1 and $2 + \omega$, respectively. Since $N(2 + \omega) = 3$, Lemma 4 applies at every place not above 3, and gives the parametrizations

$$\begin{aligned} \sigma_{17}(t) &= \left(\frac{t^2 - t + 1}{\omega t}, t \right), & \sigma_{18}(t) &= \left(-t + 2 + \omega - \frac{1}{t}, t - 1 - \omega \right), \\ \sigma_{19}(t) &= \left(\frac{t^2 + 1}{(2 + \omega)t}, t - \frac{t^2 + 1}{(2 + \omega)t} \right). \end{aligned}$$

Put $\Delta_i = 1$ for $i < 17$, and $\Delta_{17} = \omega t$, $\Delta_{18} = -t$ and $\Delta_{19} = (2 + \omega)t$. Then

$$F_i(t) = \Delta_i^4 f(\sigma_i(t)), \quad G_{ij}(t) = \Delta_i^2 H_j(\sigma_i(t))$$

are polynomials with coefficients in $\mathbb{Z}[\omega]$; the factor Δ_i^2 is enough because all H_j have total degree at most 2.

Vertex identities. For every i , let $s_i = \sum_j \rho_{ij}$, where $\rho_{ij} \in \{0, 1, 2\}$. Then

$$\Delta_i^{2s_i} W_i^3 - c_i \prod_j G_{ij}^{\rho_{ij}} = F_i Q_i, \quad 0 \leq i \leq 19, \quad (4)$$

where the polynomials W_i are listed in Table 3, and the Q_i are polynomials with coefficients in $\mathbb{Z}[\omega]$, obtained by polynomial division of the left-hand side by F_i . For example, $F_0 = -t^4 - t - 2 - 2\omega$, the neighbours of vertex 0 are 8, 13, 16 and 17, with $\rho_{0j} = 1, 1, 1, 2$, and $G_{0,8} = t - 1$, $G_{0,13} = t - 1 - \omega$, $G_{0,16} = -1 - t - t\omega$ and $G_{0,17} = 2 + 2\omega - t$. With $W_0 = 2t^3 + 4$, identity (4) at vertex 0 reads

$$\begin{aligned} (2t^3 + 4)^3 - 4(t - 1)(t - 1 - \omega)(-1 - t - t\omega)(2 + 2\omega - t)^2 \\ = (-t^4 - t - 2 - 2\omega)(-8t^5 - 40t^2 + 12t + (12t + 24)\omega). \end{aligned}$$

Separation. For every edge (i, j) with $i < j$, the norm of the resultant $\text{Res}_t(F_i, G_{ij})$ is non-zero and of the form $2^a 3^b$, see Table 4. Also the norms of all c_i are of the form $2^a 3^b$.

Proposition 5. Let (x, y) be an integer solution of (1). Then $B_p = 0$ for every prime $p \neq 2, 3$.

Proof. Let v be a place above $p \neq 2, 3$, and let k_v be its residue field. Let $(\bar{x}, \bar{y})$ be the reduction of (x, y) modulo v , and suppose that v divides H_i for some i . Then $\bar{H}_i(\bar{x}, \bar{y}) = 0$, and there is $\bar{t} \in k_v$ with $(\bar{x}, \bar{y}) = \sigma_i(\bar{t})$ and $\bar{\Delta}_i(\bar{t}) \neq 0$: for $i < 17$ we take $\bar{t} = \bar{x}$ or $\bar{t} = \bar{y}$, and for $i \geq 17$ we use Lemma 4. Since $f(x, y) = 0$, we get

$$\bar{F}_i(\bar{t}) = 0, \quad \bar{G}_{ij}(\bar{t}) = \bar{\Delta}_i(\bar{t})^2 \bar{H}_j(\bar{x}, \bar{y}) \quad \text{for all } j. \quad (5)$$

(a) If (i, j) is an edge, then v does not divide both H_i and H_j . Indeed, let $i < j$, and suppose that v divides H_i and H_j . By (5), $\bar{t}$ is a common root of $\bar{F}_i$ and $\bar{G}_{ij}$. The resultant can be

Table 3: The polynomials $W_i(t) = (P_i(t) + \omega Q_i(t))/m_i$ in the vertex identities (4).

i	m_i	$P_i(t)$	$Q_i(t)$
0	1	$2t^3 + 4$	0
1	1	$-4t - 4$	$-4t - 4$
2	1	$2t^2$	$2t$
3	1	$2t^2 + 4t - 8$	$6t - 4$
4	1	$2t^2 - 2t + 2$	$2t^2 - 2t + 2$
5	1	$2t - 2$	-2
6	1	$2t^2 - 4t + 8$	$2t^2 - 4t + 2$
7	1	1	0
8	1	$-2t^3 - 2t^2 + 4t$	$2t^3 - 4t^2 + 2t$
9	1	$4t^3 - 16t^2 + 24t - 12$	$2t^3 - 8t^2 + 12t - 6$
10	1	$2t^4 - 2t - 2$	$-2t^3 - 4t^2 - 2t - 2$
11	1	$-t + 1$	$-t^3 + t^2$
12	1	$4t^3 - 8t^2 + 8t - 4$	$2t^3 - 4t^2 + 4t - 2$
13	1	$-t^3 + 1$	$t^2 + t + 1$
14	1	2	2
15	1	$3t^3 - 2t^2 - 2t - 1$	$-4t^2 + 2t - 2$
16	1	$-t^3 - t^2 - t$	$-2t^3 - 2t^2 - 2t$
17	1	$-t^7 + t^6 - 4t^5 - t^4$ $+ 3t^3 - 4t^2 + 6t - 4$	$-t^7 + 2t^6 - 3t^5 - t^4$ $+ 2t^3 - 7t^2 + 4t - 4$
18	1	$-5t^7 + 28t^6 - 32t^5 - 16t^4$ $+ 47t^3 - 45t^2 + 18t - 1$	$-4t^7 + 41t^6 - 125t^5 + 176t^4$ $- 164t^3 + 98t^2 - 40t + 8$
19	3	$-t^7 - t^5 + 9t^4 + 18t^3$ $+ 8t^2 - 3t + 2$	$-2t^7 - 3t^6 - 2t^5 + t^2$ $- 12t - 2$

Table 4: Norms of the resultants. The entry $(i, j; a, b)$ means that the norm of $\text{Res}_t(F_i, G_{ij})$ is $2^a 3^b$.

$(i, j; a, b)$	$(i, j; a, b)$	$(i, j; a, b)$	$(i, j; a, b)$
(0, 8; 2, 1)	(2, 18; 0, 0)	(5, 18; 2, 0)	(10, 11; 0, 0)
(0, 13; 2, 0)	(2, 19; 0, 0)	(5, 19; 0, 0)	(10, 13; 8, 0)
(0, 16; 2, 1)	(3, 9; 4, 2)	(6, 8; 2, 2)	(10, 17; 4, 0)
(0, 17; 4, 2)	(3, 10; 2, 0)	(6, 9; 0, 1)	(11, 16; 0, 2)
(1, 2; 2, 0)	(3, 11; 0, 1)	(6, 10; 2, 0)	(12, 15; 0, 4)
(1, 4; 2, 0)	(3, 12; 2, 0)	(6, 18; 4, 0)	(13, 15; 2, 0)
(1, 5; 2, 0)	(3, 16; 4, 1)	(6, 19; 0, 0)	(13, 18; 2, 0)
(1, 6; 2, 0)	(3, 18; 6, 0)	(7, 11; 0, 0)	(13, 19; 0, 0)
(1, 10; 4, 0)	(4, 10; 2, 0)	(7, 17; 0, 0)	(14, 15; 4, 0)
(1, 12; 2, 0)	(4, 12; 2, 0)	(8, 9; 4, 4)	(15, 17; 0, 0)
(1, 13; 2, 0)	(4, 14; 4, 0)	(8, 16; 2, 4)	(15, 18; 4, 0)
(1, 15; 4, 0)	(4, 19; 0, 0)	(8, 17; 2, 1)	(16, 19; 2, 0)
(1, 17; 2, 0)	(5, 8; 2, 0)	(9, 12; 4, 0)	(18, 19; 2, 0)
(2, 9; 4, 0)	(5, 11; 0, 0)	(9, 14; 4, 0)	
(2, 14; 4, 0)	(5, 12; 4, 0)	(9, 15; 4, 0)	

written as $\text{Res}_t(F_i, G_{ij}) = AF_i + BG_{ij}$ with polynomials A, B with coefficients in $\mathbb{Z}[\omega]$, hence v divides $\text{Res}_t(F_i, G_{ij})$. This is impossible, because the norm of this resultant is of the form $2^a 3^b$ by Table 4.

(b) If v divides H_i , then $\bar{c}_i \prod_j \bar{H}_j^{\rho_{ij}} = \bar{W}_i(\bar{t})^3$ in k_v . Indeed, the coefficients of Q_i are in $\mathbb{Z}[\omega]$, and the coefficients of W_i have denominators dividing 3. Hence we may reduce (4) modulo v and substitute $\bar{t}$, and (5) gives

$$\bar{\Delta}_i(\bar{t})^{2s_i} \left(\bar{W}_i(\bar{t})^3 - \bar{c}_i \prod_j \bar{H}_j^{\rho_{ij}} \right) = 0.$$

Now consider the sum (3). All c_i are units at v , because their norms are not divisible by p . By (H2), a term $e_{ij}\beta_v(H_i, H_j)$ in which both H_i and H_j are units vanishes, and so does a term $\beta_v(H_i, c_i)$ in which H_i is a unit. By (a), every remaining term involves exactly one value H_i divisible by v . For such an i , we collect all these terms, using (H1) and $\rho_{ij} = -e_{ji}$ for $j < i$:

$$\sum_{j>i} e_{ij}\beta_v(H_i, H_j) + \sum_{j<i} e_{ji}\beta_v(H_j, H_i) + \beta_v(H_i, c_i) = \beta_v\left(H_i, c_i \prod_j H_j^{\rho_{ij}}\right).$$

By (a), the second argument is a unit, and by (b) its reduction is a non-zero cube in k_v , so that its value of χ is 0. By (H2), this symbol is 0. Hence $B_v = 0$ for every place v above p , and $B_p = 0$. $\square$

5 The primes 2 and 3

Let $p \in \{2, 3\}$. We split the hypothetical integer solutions into finitely many residue classes $x \equiv a, y \equiv b \pmod{p^k}$, and compute B_p in each class.

Lemma 6. *Let $k \geq 1$, and let $f(a, b) \equiv 0 \pmod{p^k}$. Every integer solution of (1) with $x \equiv a, y \equiv b \pmod{p^k}$ satisfies $x \equiv a + p^k s, y \equiv b + p^k t \pmod{p^{k+1}}$ for some $0 \leq s, t < p$ with $f(a + p^k s, b + p^k t) \equiv 0 \pmod{p^{k+1}}$.*

Proof. The digits s and t of x and y in base p are determined by x and y , and $f(x, y) = 0$ is divisible by p^{k+1} . $\square$

Classes of the auxiliary values. Let (x, y) be an integer solution with $x \equiv a, y \equiv b \pmod{p^k}$. Then $H_i(x, y) \equiv H_i(a, b)$ modulo $p^k \mathbb{Z}[\omega]$, and this determines a set of possible classes of $H_i(x, y)$ modulo cubes at the place above p , as follows.

- $p = 2$. If $2^k \nmid H_i(a, b)$, then $e = v(H_i(x, y)) = v(H_i(a, b)) < k$, and the unit part of $H_i(x, y)$ is known modulo 2^{k-e} , hence modulo 2, so that the class $(e \bmod 3, \chi)$ is determined. If $2^k \mid H_i(a, b)$, we allow all classes, with $\chi = 0$ if H_i has rational coefficients.
- $p = 3$. Let $e = v(H_i(a, b))$, the valuation at π . If $e < 2k$, then $v(H_i(x, y)) = e$, and the unit part is known modulo π^{2k-e} . If $2k - e \geq 4$, its class is determined by Lemma 2, and otherwise we allow the classes of all units modulo 9 which are congruent to it modulo π^{2k-e} . If $e \geq 2k$, we allow all classes. If H_i has rational coefficients, we use instead that $H_i(x, y)$ is a rational integer: if $3^k \nmid H_i(a, b)$, its 3-adic valuation j and its unit part modulo 3^{k-j} are known, and we allow the classes of $3^j u$ for all rational units u modulo 9 compatible with it; if $3^k \mid H_i(a, b)$, we allow the classes of $3^j u$ for all $j \in \{0, 1, 2\}$ and all units u modulo 9.

In every case, the allowed set contains the class of $H_i(x, y)$. The constants c_i have fixed classes.

The computation. We start with the solutions of (1) modulo p , and refine a residue class by Lemma 6 as long as (3), computed with (2) for $p = 2$ and with Lemma 2 for $p = 3$, takes more than one value when the classes of the H_i run over the allowed sets. Modulo 2, the solutions are $(x, y) \equiv (0, 0)$ and $(1, 0)$. The class $(1, 0)$ is terminal; the class $(0, 0)$ lifts to $(2, 0)$ and $(0, 2)$ modulo 4, the first of these is terminal, and the second lifts to $(0, 6)$ and $(4, 2)$ modulo 8. Modulo 3, the only solution is $(1, 1)$, which lifts to $(1, 7)$, $(4, 1)$ and $(7, 4)$ modulo 9, and then to the nine classes modulo 27 in Table 5. By Lemma 6, every integer solution lies in one of these 13 terminal classes.

Table 5: The terminal residue classes $x \equiv a$, $y \equiv b \pmod{p^k}$ and the values of B_p . The last column lists the H_i whose class is not determined by the residue class. In every class, B_p takes the given value for all allowed choices of these classes.

p	k	a	b	B_p	Undetermined
2	1	1	0	2	H_4, H_{12}
2	2	2	0	2	–
2	3	0	6	2	H_2
2	3	4	2	2	–
3	3	1	7	0	–
3	3	4	10	0	H_{11}
3	3	7	22	0	H_{16}
3	3	10	16	0	–
3	3	13	19	0	H_{11}
3	3	16	4	0	H_{16}
3	3	19	25	0	–
3	3	22	1	0	H_{11}
3	3	25	13	0	H_{16}

In every terminal class, at most two of the H_i have more than one allowed class, and there are at most 9 choices. For example, in the class $x \equiv 1$, $y \equiv 0 \pmod{2}$, the classes (e, χ) of the values at the place above 2 are

i	0	1	2	3	4	5	6	7	8	9
H_i	$(0, 2)$	$(0, 2)$	$(0, 0)$	$(0, 1)$	$(e_4, 0)$	$(0, 2)$	$(0, 2)$	$(0, 1)$	$(0, 2)$	$(0, 0)$
c_i	$(2, 0)$	$(2, 2)$	$(2, 1)$	$(1, 1)$	$(2, 0)$	$(1, 1)$	$(1, 0)$	$(0, 0)$	$(2, 0)$	$(1, 0)$
i	10	11	12	13	14	15	16	17	18	19
H_i	$(0, 2)$	$(0, 0)$	$(e_{12}, 0)$	$(0, 0)$	$(0, 1)$	$(0, 2)$	$(0, 0)$	$(0, 0)$	$(0, 0)$	$(0, 1)$
c_i	$(1, 0)$	$(0, 1)$	$(2, 0)$	$(0, 1)$	$(1, 1)$	$(1, 2)$	$(1, 1)$	$(0, 0)$	$(0, 2)$	$(0, 2)$

where e_4 and e_{12} are the unknown valuations of $H_4 = x + 1$ and $H_{12} = x - y + 1$ modulo 3. By (2), the terms of (3) which contain e_4 add up to e_4 times

$$\sum_j \rho_{4j} \chi(H_j) + \chi(c_4) = 2 \cdot 2 + 2 \cdot 2 + 1 \cdot 0 + 2 \cdot 1 + 2 \cdot 1 + 0 = 12 \equiv 0,$$

where the neighbours of vertex 4 are 1, 10, 12, 14, 19; similarly the terms which contain e_{12} cancel, and (3) gives $B_2 = 2$.

The result of the computation is

$$B_2 = 2, \quad B_3 = 0 \tag{6}$$

at every integer solution of (1).

6 End of the proof

Suppose that (x, y) is an integer solution of (1). By Section 3, all values $H_i(x, y)$ are non-zero. By Proposition 5 and (6),

$$\sum_p B_p = B_2 + B_3 = 2 \neq 0 \quad \text{in } \mathbb{Z}/3\mathbb{Z},$$

which contradicts Proposition 3. Hence equation (1) has no integer solutions. $\square$

References

- [1] J. Neukirch, *Algebraic Number Theory*, Grundlehren der mathematischen Wissenschaften 322, Springer, Berlin, 1999.